



Capstone Project Final Analysis Report

PGC in Data Analytics for Business Decision Making

Multimedia Content Analysis for Streaming Services Industry

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Executive Summary

The client company have requested the assistance in understanding a system of measurement of producing high grossing movies or tv series, as well as leveraging these same insights for streaming services platforms. Technology has introduced new ways of entertainment and people have shifted from cable television to streaming platforms for many reasons such as convenience, less expensive, on demand broadcasting, etc. They provide scalable personalized content for different demographics for a reasonable subscription fee. Netflix, Disney Plus, HBO Max, Prime Video, Hulu are few of the top streaming services. This new means of entertainment has paved ways for new means of investment as well. Earlier production houses earned revenue only by movie ticket sales in cinemas, but now they have a new platform where they can sell their content and reach new levels of profitability and success.

This report will explore these opportunities for different stakeholder categories, supported by data obtained from Netflix platform from 1990 to 2021. Data of 3481 movies and TV series, over the last 30 years (1990 to 2021) were gathered, studied, and analysed. It contains details of the respective genres, tags, languages, Actors, IMDb score, Hidden gem score, IMDb ratings, Box office, Release date, Netflix release date, Awards nominated, Awards received, etc. These data have been analysed to derive useful insights of the report aligned to with the following business opportunities, identifying:

- What type of content (movie/TV series) is mostly likely to be profitable?
- Which genres generate the highest incomes?
- Which translated/dubbed languages have the potential to generate higher earnings?
- Development and evaluation of data analytics models to determine various aspects of the content (movie/TV series), such as IMDb score, box office collection, awards received, etc.
- What factors supports content to receive recognition (awards)?
- What is the most rewarding period to release content in OTT platforms?

Visual displays were created to analyse past data and different algorithms were used to build prediction models which has the capacity to help the client invest in commercially successful and critically acclaimed content. Although the performance of some models was questionable, they can be put to deployment with improvements using more quality data, feature selection, optimization techniques, etc. Final recommendations structured around a standard criterion organized at the revenue and recognition output levels of content, were provided to the client.

1. Problem Description

Multimedia content has become an indispensable part of life for many people. They have a major effect on real life, but also on people's personal development. Considering that nearly almost all the people watch content on devices other than an actual TV, this was a huge advantage for OTT, and as a result, OTT platforms now conquer the media industry. For investors, investing on content may sound glam, however in the truth it can be very complex and a very risky endeavor. Exploring the right genre for the target audience, knowing the most profitable time to produce content, knowing if the content is award worthy, are just only a few of the difficulties investors always bargain through. A successful movie can open more possibilities for all involved, and likewise a failed movie can cause nightmares for all involved. The key to success is understanding the needs of the audience leveraging data analysis.

The following business goals will be met through this analysis:

- To determine which kind of content can bring more revenue
- To understand attributes (genre, language, country, content type, etc.) needed to:
 - Attain commercial success
 - Achieve critical acclamation
- To uncover the most rewarding time of the year to release content
- To provide a standard criterion for decision making in the future

The following data analysis goals are achieved to cater the given business goals:

- Exhibit important details of a movie/TV series in an interactive dashboard
- Revenue analysis of multimedia content
 - Annual box office of movies and TV series from 1990 to 2021
 - Box office variation by month
 - Highest earning production houses in the past 30 years
 - Top 5 highest box office movies/TV series
 - Top 5 categories of highest box office genre and languages
- Analysis of recognition/awards received by content
 - Variation of content released, and awards received over time
 - Top award-winning genre categories
 - Top award-winning production houses
- Identify correlation between different characteristics of a movie/TV series
- Identify factors impacting the IMDb score, verdict and box office of movies and TV series and their degree of impact using data models
 - Regression models to predict the IMDb score of movies and TV series
 - Naïve Bayes Model and Support Vector Model to accurately predict the verdict of a movie or TV series.

2. Data Description

Dataset used for the analysis has information on movies and tv series released on Netflix platform compiled from the year 2015 to 2021. This dataset consists of movies and tv series which were originally released from 1990 to 2021. There is no private information used in this dataset and all data are publicly available on the website. The raw dataset contained 9,425 rows and 31 columns of data. Some of the main attributes of this dataset are, movie name, Date of release, Box office collection, IMDB Collection, Hidden gem score, IMDB votes, etc. Refer *Appendix C* for first 5 rows of the original dataset.

Dataset statistics are as follows:

Dataset statistics		Variable types	
Number of variables	31	Unsupported	4
Number of observations	9425	Categorical	17
Missing cells	54684	Numeric	8
Missing cells (%)	18.7%	DateTime	2
Duplicate rows	0		
Duplicate rows (%)	0.0%		
Total size in memory	2.2 MiB		
Average record size in memory	248.0 B		

Figure 1: Dataset Statistics

All the records were thoroughly studied and the attributes providing no input in the analysis were dropped from the table – this includes any empty, null, and duplicate records of data. Where necessary for this analysis, additional data columns were also be added prior to performing the analysis.

3. Data Preparation Details

The preparation of our data consisted of several parts: parsing, transforming, and merging. Python Pandas was used to perform many tasks like generating the profile report, create data frames to create visualizations.

The first steps of the data cleansing consisted of deconstructing the dataset to standardize the columns and generate the most important variables. After deconstructing the movie dataset, the new data frame included the following columns: IMDB Votes, Box Office, Month of release, Hidden gem score, IMDB Votes, Movie Name, Genre, No. of IMDB votes, etc.

Second step was to transform the dataset to ensure the variables were standardized. To be able to recommend the movies or series' verdict, a column was created to assign categories "Blockbuster, Super Hit, Hit, Flop". We have assigned verdict based on IMDb scores, instead of box office revenue, reason being the ultimate goal is viewer satisfaction.

The cleansed dataset amounted to 14 columns and 3754 records of data.

Title	Genre	Languages	Series or Movie	Hidden Ger Country Availability	Runtime	IMDb Score	Boxoffice	Release Date	Netflix Release Date	IMDb Votes	Year	Verdict
0 Lets Fight Ghost	Crime, Drama, Fanti	Swedish, Spanish	Series	4.3 Thailand	< 30 minutes	7.9	2122065	2008-12-12	2021-03-04	205926	2008	Superhit
1 HOW TO BUILD A GIRL	Comedy	English	Movie	7 Canada	1-2 hour	5.8	70632	2020-05-08	2021-03-04	2838	2020	Average
6 The Invisible	Crime, Drama, Fanti	English	Movie	2 Lithuania,Poland,Fra	1-2 hour	6.2	20578909	2007-04-27	2021-03-03	34738	2007	Hit
9 Joker	Crime, Drama, Thrill	English	Movie	3.5 Lithuania,Poland,Fra	1-2 hour	8.4	335451311	2019-10-04	2021-03-03	951938	2019	Superhit
10 I	Action, Adventure, I	English, Sanskrit	Movie	2.8 Lithuania,Poland,Fra	1-2 hour	6.5	474544677	1999-05-19	2021-03-03	733336	1999	Hit
11 Harrys Daughters	Adventure, Drama, I	English	Movie	4.4 Lithuania,Poland,Fra	1-2 hour	8.1	381409310	2011-07-15	2021-03-03	766594	2011	Superhit
16 Comrades: Almost a Love Sto	Drama, Romance	Cantonese, Man	Movie	6.7 South Korea	1-2 hour	8.1	17676	1996-11-02	2021-03-01	5047	1996	Superhit
17 The Closet	Comedy	French	Movie	3.8 South Korea	1-2 hour	7	6678894	2001-08-10	2021-03-01	16657	2001	Hit
18 The Mysterians	Action, Family, Sci-F	Japanese, English	Movie	7.1 Japan	1-2 hour	6.2	975000	1959-05-15	2021-03-01	1383	1959	Hit
26 Trial by Fire	Biography, Drama	English	Movie	6.2 United Kingdom	> 2 hrs	6.9	148504	2019-05-17	2021-03-01	3657	2019	Hit
34 Framing John DeLorean	Documentary, Biogr	English	Movie	8 Poland	1-2 hour	6.5	145625	2019-06-07	2021-03-01	1398	2019	Hit
36 Alice	Comedy, Romance	English, Cantone	Series	3.5 Lithuania,Poland,Fra	< 30 minutes	6.6	7331647	1991-01-10	2021-03-01	13379	1991	Hit
39 Ordinary People	Drama	English	Movie	4.2 South Korea	1-2 hour	7.7	54766923	1980-09-19	2021-02-28	47265	1980	Superhit
40 Paths of the Soul	Drama	Tibetan	Movie	8.9 South Korea	1-2 hour	7.5	31747	2017-06-20	2021-02-28	738	2017	Superhit
42 Rebel in the Rye	Biography, Drama, I	English	Movie	3.1 South Korea	1-2 hour	6.7	378294	2017-09-15	2021-02-28	8472	2017	Hit
43 The Return	Drama	Russian	Movie	4.1 South Korea	1-2 hour	8	504256	2003-06-25	2021-02-28	42530	2003	Superhit
46 Stand by Me	Adventure, Drama	English	Movie	4.1 South Korea	1-2 hour	8.1	52287414	1986-08-22	2021-02-28	364302	1986	Superhit
47 Wonderstruck	Adventure, Drama, I	English, Spanish	Movie	3.6 South Korea	1-2 hour	6.2	1060377	2017-10-20	2021-02-28	9533	2017	Hit
48 Keys To The Heart	Comedy, Drama, Fa	Korean	Movie	8.4 South Korea	> 2 hrs	7.5	75134	2018-01-26	2021-02-28	825	2018	Superhit
49 Intimate Strangers	Drama, Romance, T	French	Movie	6.4 South Korea	1-2 hour	7	2110589	2004-09-03	2021-02-28	4813	2004	Hit
52 Gonjiam: Haunted Asylum	Horror	Korean	Movie	6.3 South Korea	1-2 hour	6.3	115252	2018-03-28	2021-02-28	5195	2018	Hit
54 Extreme Job	Action, Comedy, Cri	Korean, Mandari	Movie	5.4 South Korea	1-2 hour	7.1	1565885	2019-06-21	2021-02-28	6757	2019	Hit
55 Default	Drama	Korean, English	Movie	8.1 South Korea	1-2 hour	6.6	203775	2018-11-28	2021-02-28	1007	2018	Hit
58 The Accidental Detective 2: In	Comedy, Crime, Thr	Korean	Movie	8.2 South Korea	1-2 hour	6.5	179045	2018-06-22	2021-02-28	513	2018	Hit

Figure 2: Cleansed Dataset Preview

After the initial analysis of the data, it was determined that our analysis would have more impact if we treated movies and series separately, as both have its own characteristics and to increase the accuracy of our predictions and recommendations.

4. Data Analysis Solution

4.1. Dashboard

4.1.1. Dashboard 1: Movie Content Details

Dashboard 1 displaying details of movies to give an overview on an individual basis.

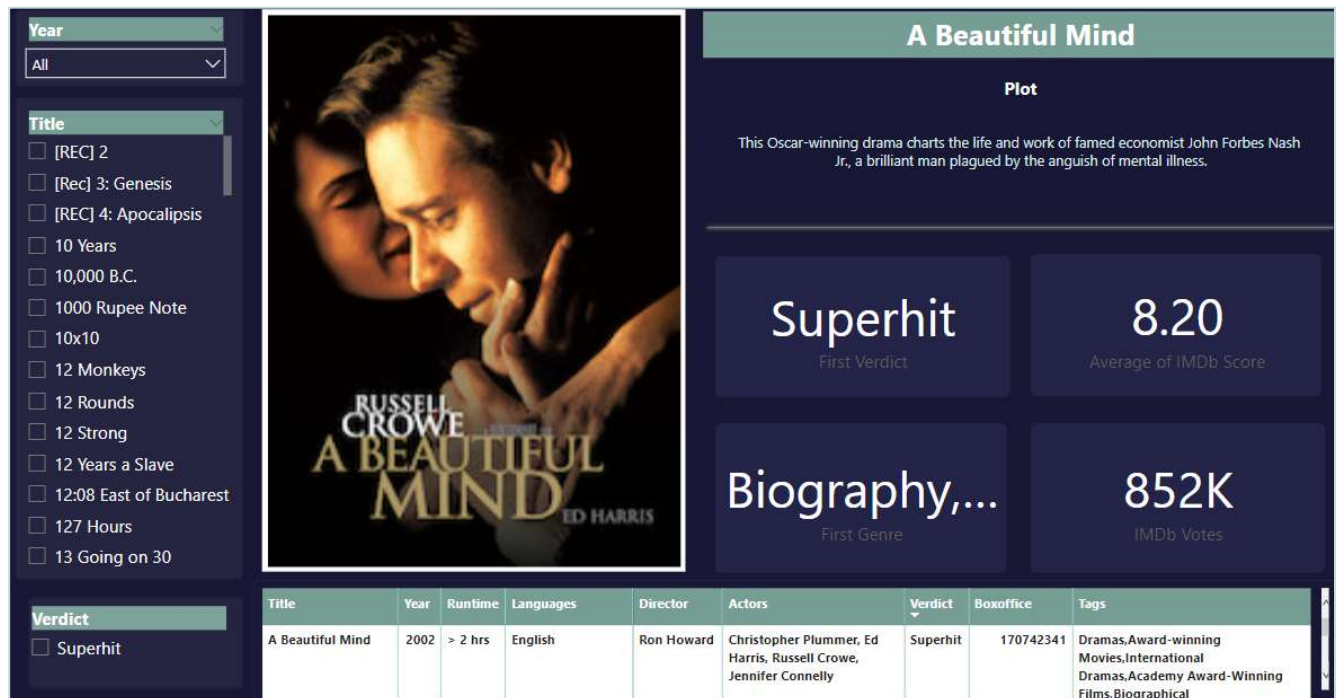


Figure 33: Movie content Details

4.1.2. Dashboard 2: Revenue Analysis of Content

Dashboard 2 provides all attributes of both movies and TV series in terms of revenue.

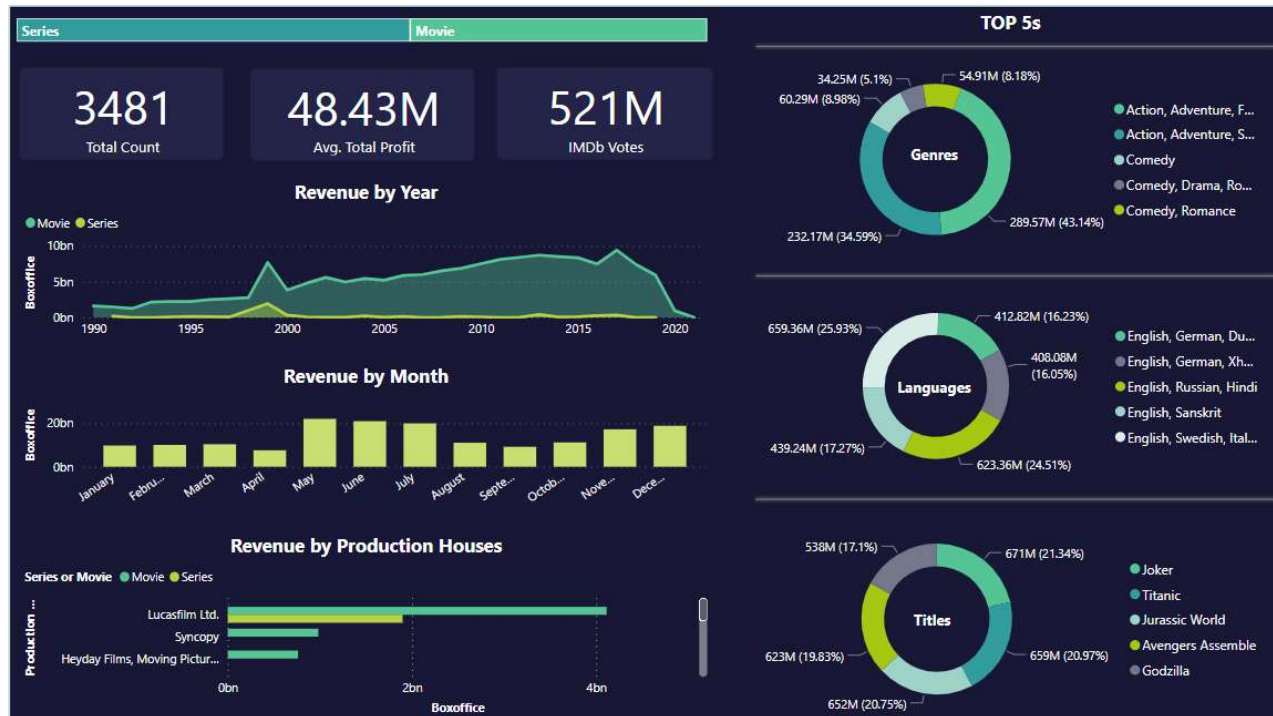


Figure 4: Revenue Dashboard

Insights

- Movie shows a higher generation of revenue throughout the years of 1990 to 2021, as opposed to TV Series. Both movies and TV series show a peak in revenue in the year 1999. The year 1999 was called the “Year of the movies” as it was said to be the start of a new age for cinema, with many released of original stories. (Independent, 2019)
- Our findings show that highest number of contents released had taken place in the month of October (*Appendix A*). However, the total revenue generated throughout years 1990 to 2021, content released on the month of October does not show much revenue generated. Instead, the months of May, June, July, November, and December, displays the highest revenue earned.
- We found there are approximately 1513 combinations of genres to date. As per the analysis, the following are the top 5 genres combinations with regard to having had generated the highest revenue:

No.	Genre Combinations	Revenue	Movie Examples
1	Action, Adventure, Fantasy, Sci-Fi	\$289.57M	Doctor Strange, Aquaman, Suicide Squad
2	Action, Adventure, Sci-Fi	\$23.17M	2012, After Earth, Edge of Tomorrow
3	Comedy	\$60.29M	Office Space, The Hangover,
4	Comedy, Drama, Romance	\$34.25M	Love Actually, The Artist, 50/50
5	Comedy, Romance	\$54.91M	27 Dresses, About Last Night, Leap Year

Recurring genres: Action, Adventure, Science Fiction, Comedy, Romance

- The following are the top 5 languages that displayed to have generated the highest revenue:

No.	Languages	Revenue
1	English, Swedish, Italian, French	\$659.36M
2	English, Russian, Hindi	\$623.36M
3	English, Sanskrit	\$439.23M
4	English, German, Dutch, French, Spanish, Chinese, Greek, Indian	\$412.82M
5	English, German, Xhosa, Russian, Romanian, Hindi	\$408.08M

Recurring languages apart from English: Hindi, French, German/Dutch

- Top 5 movies with the highest revenue generated, through 1990 to 2021 (30 years)

No.	Movie Title	Released Year	Revenue	Verdict
1	Joker	2019	\$671M	Superhit
2	Titanic	1997	\$659M	Superhit
3	Jurassic World	2015	\$652M	Hit
4	Avengers Assemble	2012	\$623M	Superhit
5	Godzilla	1998	\$538M	Average

- Top 5 studios with the highest revenue generation, through 1990 to 2021 (30 years)

No.	Production House	Revenue
1	Lucasfilm Ltd.	\$6B
2	Syncopy	\$983M
3	Heyday Films, Moving Picture Company, Warner Bros.	\$763M
4	20th Century Fox, Lightstorm Entertainment, Paramount Pictures	\$659M
5	Amblin Entertainment, Universal Pictures, Legendary Pictures	\$652M

4.1.3. Dashboard 3: Analysis of recognition/awards received by content

Dashboard 3 provides all attributes of both movie and TV series in terms of recognition.

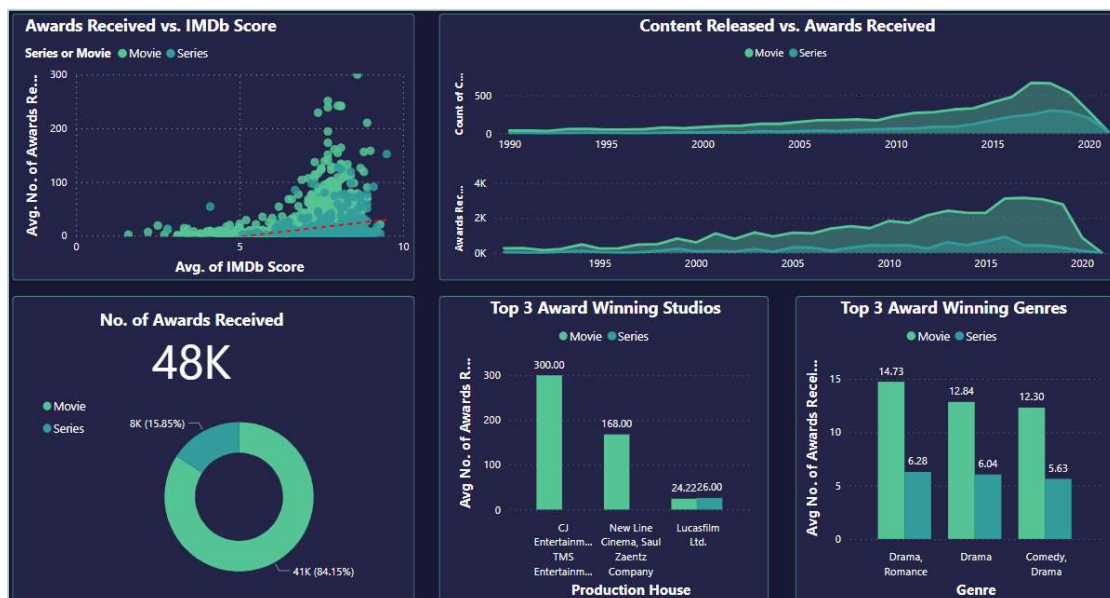


Figure 5: Recognition/Awards Received

Insights:

- IMDb score and awards received has a slightly positive correlation. If a movie/series has a high IMDb score, most likely it will win awards/recognition.
- Awards Received for Movie (40636) was higher than Series (7655). Awards received for movies amounts to 5.3 times the awards received for tv series.
- Top 3 award-winning studios

No.	Production House	Avg. no. of Awards	Content Type
1	CJ Entertainment, TMS Entertainment	300	Movie
2	New Line Cinema, Saul Zaentz Company	168	Movie
3	Lucasfilm Ltd.	26	Series
4	Lucasfilm Ltd.	24	Movie

- Top 3 award-winning genres: Drama, Romance, Comedy

No.	Production House	Avg. no. of Awards	Content Type
1	Drama, Romance	15	Movie
2	Drama	13	Movie
3	Comedy, Drama	12	Movie
4	Drama, Romance	6	Series

4.2 Models

4.2.1. Pandas Profile Reporting

Identification of correlation between different characteristics of a movie/tv series

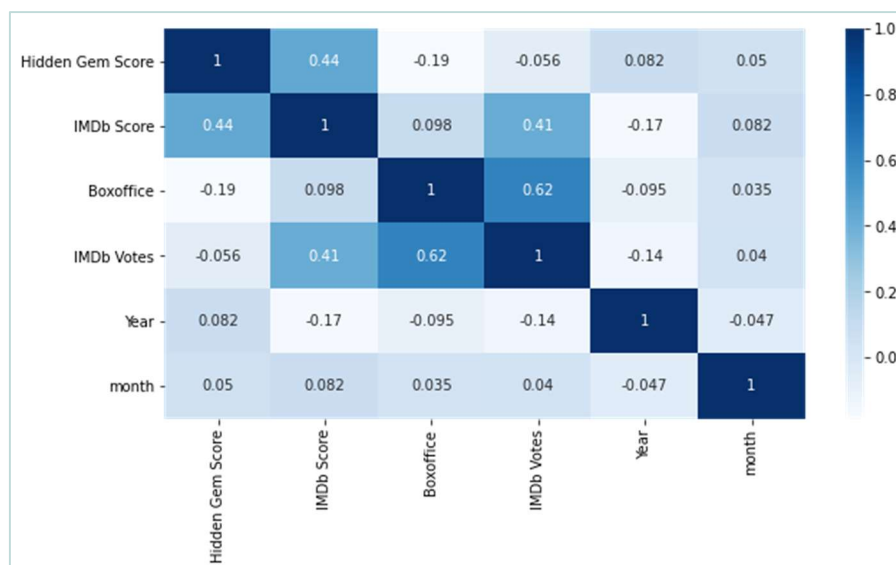


Figure 6: Correlation Heatmap 1

- Features IMDb Votes and Box Office are highly correlated.

- IMDb votes and IMDb score are also slightly correlated.

Identification of correlation between different scores and votes

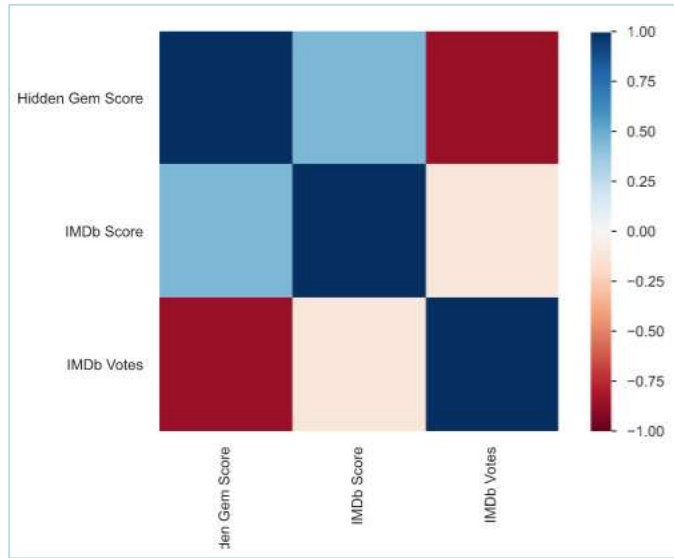


Figure 7: Correlation Heatmap 2

- IMDb votes and IMDb scores are positively correlated
- IMDb votes and Hidden Gem scores have strong negative correlation which implies that although a content is highly viewed, it does not ensure the quality of the content, instead they are negatively correlated.

4.2.2. Regression Model

Regression model was used to build a model to identify the variables impacting the verdict of a movie.

Model 1: Regression model to predict the IMDb score/verdict of Movies

Input Variables: IMDb votes, Hidden Gem score, Box Office

Output Variable: IMDb score (Verdict is derived from IMDb score intervals)

Model Outcomes:

SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.634463963								
R Square	0.402544521								
Adjusted R Square	0.402055871								
Standard Error	0.706189984								
Observations	3672								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	3	1232.482625	410.82754	823.7899	0				
Residual	3668	1829.24735	0.4987043						
Total	3671	3061.729975							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	5.503065664	0.031665363	173.78817	0	5.440982206	5.565149	5.440982	5.565149122	
Hidden Gem Score	0.257191926	0.007512173	34.236689	3.6E-223	0.242463478	0.27192	0.242463	0.271920374	
IMDb Votes	2.35202E-06	7.27667E-08	32.322665	6.7E-202	2.20935E-06	2.49E-06	2.21E-06	2.49468E-06	
Boxoffice	-1.81467E-09	2.12036E-10	-8.5583135	1.66E-17	-2.23039E-09	-1.4E-09	-2.2E-09	-1.39895E-09	

Figure 8: Regression Model 1

Performance Evaluation:

By looking at the Adjusted R squared value for the model, which is slightly higher than 0.4, we can conclude that the model is not a very good fit but not too bad either. All p-values of independent variables are smaller in value than the confidence level 0.05 which is a good indication of the model as it signified that all independent variables are statistically significant to the model.

Model 2: Regression model to predict the IMDb score/verdict of TV series

Input Variables: IMDb votes, Hidden Gem score, Box Office

Output Variable: IMDb score of the TV series

Model Outcomes:

SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.530739021								
R Square	0.281683909								
Adjusted R Square	0.263498691								
Standard Error	0.617053652								
Observations	82								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	2	11.79558238	5.8977912	15.489719	2.11089E-06				
Residual	79	30.07966153	0.3807552						
Total	81	41.8752439							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	6.594875872	0.084681226	77.878842	1.848E-76	6.426322108	6.7634296	6.426322108	6.763429637	
X Variable 1	2.93415E-06	5.27297E-07	5.5644992	3.476E-07	1.88459E-06	3.984E-06	1.88459E-06	3.9837E-06	
X Variable 2	-4.35282E-09	9.7784E-10	-4.4514692	2.767E-05	-6.2992E-09	-2.406E-09	-6.2992E-09	-2.40648E-09	

Figure 9: Regression Model 2

Performance Evaluation:

Model is not a good fit as per the Adjusted R squared value (26%). The reason for this can be the limited data on TV series in the chosen dataset. This makes us wonder if the verdict of a TV series is impacted by variables which are not selected to build the model and are different from what impacts the verdict of a movie.

4.2.3. Naïve Bayes Model

Naïve Bayes Model can be used to identify the verdict of the movie and series

Following is the model:

Input Variables: IMDb Scores, Hidden Gem Score

Output Variables: Verdict

Estimator: Naive Bayes - GB				
[[125 0 1 12 0]				
[0 1 0 0 0]				
[0 0 13 0 0]				
[28 0 0 415 11]				
[0 1 0 28 116]]				
	precision	recall	f1-score	support
Blockbuster	0.82	0.91	0.86	138
Superhit	0.50	1.00	0.67	1
Hit	0.93	1.00	0.96	13
Average	0.91	0.91	0.91	454
Flop	0.91	0.80	0.85	145
accuracy			0.89	751
macro avg	0.81	0.92	0.85	751
weighted avg	0.89	0.89	0.89	751

Figure 10: Naive Bayes Classification Report

Performance Evaluation:

There are total of 751 data points from which the accuracy of the model is 89%. The recall value of macro average is 92% and Weighted average is at 89%. The model is not that accurate. So, we will perform SVM

4.2.4. SVM Model

Here, we need to perform SVM to predict the Verdict and give more accurate results.

Estimator: SVM				
[[135 0 0 3 0]				
[0 0 0 0 1]				
[2 0 11 0 0]				
[1 0 0 448 5]				
[0 0 0 2 143]]				
	precision	recall	f1-score	support
Average	0.98	0.98	0.98	138
Blockbuster	0.00	0.00	0.00	1
Flop	1.00	0.85	0.92	13
Hit	0.99	0.99	0.99	454
Superhit	0.96	0.99	0.97	145
accuracy			0.98	751
macro avg	0.79	0.76	0.77	751
weighted avg	0.98	0.98	0.98	751

Figure 11: SVM Classification Report

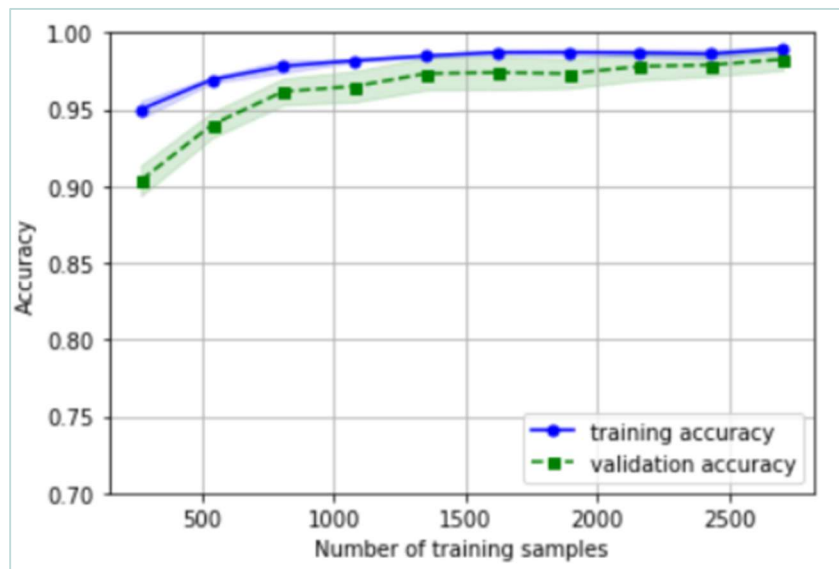


Figure 12: SVM Learning Curve

Performance Evaluation:

After performing the SVM model, we can see that this model gives 98% accuracy with the same 751 data points. The overall precision, recall, and f1 values are more accurate than Naïve Bayes.

5. Conclusions

5.1 Operational recommendations

- There are more movie data than TV series, therefore the analysis could be biased towards movies.
- Regression models 1 & 2
 - Considering the performance metrics, the models' performance was relatively weak.
 - Models were built for movie and series separately. When comparing the two, it was found that the regression model for predicting movie IMDb score performed better than the model for predicting for TV series.
 - Limitations for this under-performance are not having a balanced dataset, lack of adequate suitable independent variables, and correlations between variables.
- Naïve Bayes and SVM Model
 - In the Netflix dataset, we had incomplete data. After our data preparation, we tried to create a model which would help us in accurately categorizing a movie or series into Blockbuster, Superhit, Hit, Average and Flop.

- Few limitations faced while creating the model are high correlation between the independent variables and not having the right data may result in faulty evaluations. The independent variable would need to be checked and rectified to obtain with the accurate results.
- Therefore, predictive analytics can be used once a movie has been created to predict its verdict, revenue, and possibility of winning an award.
- For this Netflix dataset, we recommend that investors use descriptive analytics thoroughly as much as predictive analytics for their decision-making process. Using descriptive analytics of data of 3 decades, the client is provided with **historically proven insights and recommendations**. Descriptive analytics in this case can help as much as or even more so than predictive analytics, advise investors decide on many aspects when creating content or investing in them.
- Furthermore, the data used did not have data on customer demographic, customer view time and count, social shares, and conversion rates, movie budget, to name a few. With inclusivity of such data, investors can use both descriptive and predictive analytics to understand how to also target their ideal target market better.
- Although the performance of the predictive models was questionable, they can be put to deployment with improvements using more quality data and different techniques.

5.2 Recommendations to the client

The above findings remind us that the multimedia industry is a vast area and is evolving at a very fast pace. The data analysis in this report yields the following major conclusions for both production houses and OTT Platforms, that will hold a base for future content creation efforts for better returns, based off the most beneficial content (movie and TV series) from data of over 3000 records over the last 30 years.

- **Produce both movie and TV series**
Observing the production houses with the highest revenue earned, top production houses like Lucasfilm Ltd earns billions of dollars by combining different content types, as opposed to production houses that produce either content type. Lucasfilm Ltd also stands the 3rd award-winning production house. Therefore, it is highly recommended to produce both types of content.
- **Invest more in movie production than in TV series production**
Over the last 3 decades, movie content has raked in as much as billions in profits earning money from box office revenue. Moreover, movies have got more recognition than TV series has, accounting for about 84% of the awards received. It is highly recommended that the client, does opt to have both types of content but invest more on movies, in terms of gaining a higher revenue and having a higher chance in gaining recognition.
- **Create content with Action, Adventure, Science Fiction, Comedy, and/or Romance genres**
 - **Combination genres:** the type of content that falls in this category are content that continually bring in creativity, that cultivates customers' curiosity, that relies heavily on computer-generated effects, and has different emotional tones.
 - **Single genre:** the genre Comedy stands out to be the 3rd most revenue generating genre as a single genre among the combinations without any dependencies of other genres. A single genre having the ability to generate revenue on its own can be considered the “golden egg” of the industry for investor to use it as the main genre or with other supporting genres. The genre comedy is also the

3rd award-winning genre based on the analysis. Moreover, comedy content is said to earn higher profits with low budgets. (Follows.S, & Nash. B, 2021)

- **Release content in May, June, July, November, and December**

Over the last 3 decades, months with the highest grossing are May to July, November, and December. “Dump Months” or the months with the lowest grossing were January to April, and August to September, therefore it is recommended the client refrain from releasing content during this period.

- **Prioritize translate/dubbing of content in Hindi, French, and German/Dutch languages**

New languages mean the potential to tap into international markets, which brings new geographical revenue streams. As per our analysis, apart from the English language, content in languages of Hindi, French, and German/Dutch, show highest revenue. It is recommended that investors may not necessarily have to dub these three languages per se but do try to dub their content in different languages to encourage localisation in addition to revenue generation.

Its important to note that OTT Platforms behavior is influenced by external factors such as politics, societal concerns, cultural changes, etc. therefore predictability may most likely not always be accurate.

It is an incredibly exciting time to be in the multimedia industry due to the technological advancement happening with mere “convenience”. Given the above recommendations, it is vital for the client to remember that multimedia industry changes very fast, and new development in media industry will shape and change consumer attitude towards multimedia content.

5. References

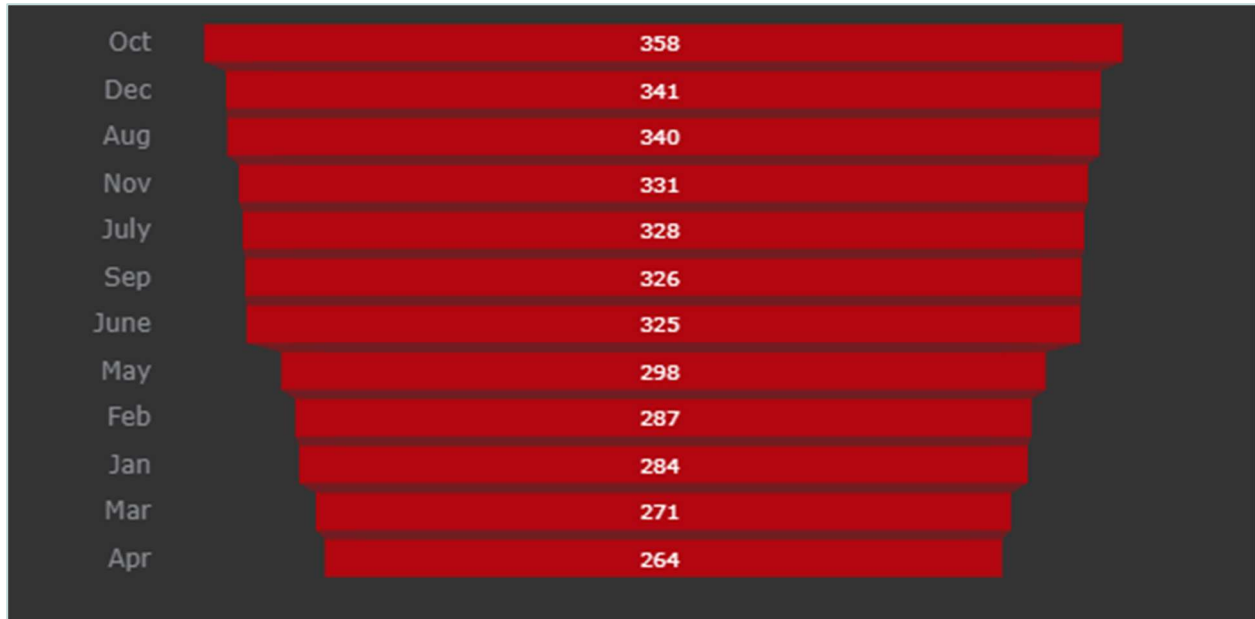
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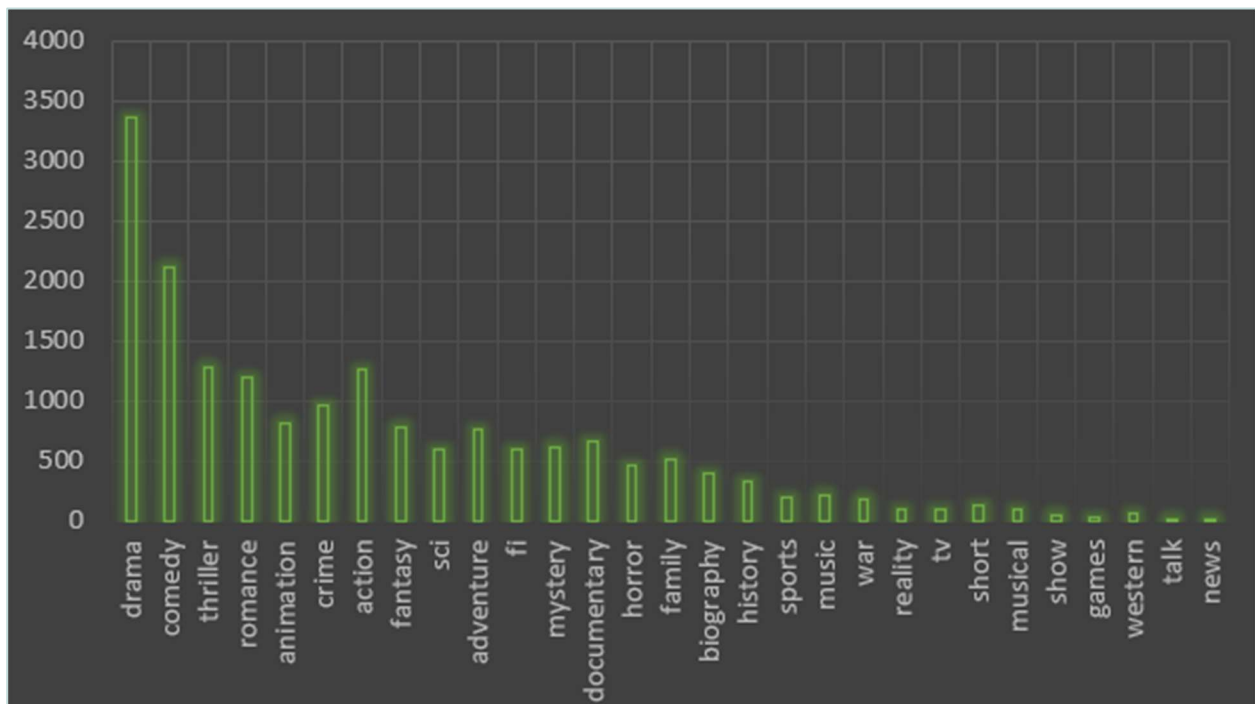
Dataset source: <https://www.kaggle.com/syedmubarak/netflix-dataset-latest-2021>

6. Appendix

Appendix A: Number of contents released by month



Appendix B: Number of contents released by genre



Appendix C: First 5 rows of the original dataset

	Title	Genre	Tags	Languages	Series or Movie	Hidden Gem Score	Country Availability	Runtime	Director	
0	Lets Fight Ghost	Crime, Drama, Fantasy, Horror, Romance	Comedy Programmes, Romantic Comedies, Horror ...	Swedish, Spanish	Series	4.3	Thailand	< 30 minutes	Tomas Alfredson	John Li
1	HOW TO BUILD A GIRL	Comedy	Dramas, Comedies, Films Based on Books, British	English	Movie	7.0	Canada	1-2 hour	Coly Giedroyc	Caitlin
2	The Con-Heartist	Comedy, Romance	Comedies, Comedies, Romantic Films, Thai...	Thai	Movie	8.6	Thailand	> 2 hrs	Mez Tharatorn	Patt Bhiboons Mez Thu
3	Gleboka woda	Drama	TV Dramas, Polish TV Shows, Social Issue TV Dramas	Polish	Series	8.7	Poland	< 30 minutes	NaN	
4	Only a Mother	Drama	Dramas, Dramas, Social Issue Movies Based on Boo...	Swedish	Movie	8.3	Lithuania, Poland, France, Italy, Spain, Greece, Bel...	1-2 hour	Alf Sjöberg	Joh

Writer	Netflix Link	IMDb Link	Summary	IMDb Votes	Image	Poster	TMDb Trailer
John Ajvide Lindqvist	https://www.netflix.com/watch/81415947	https://www.imdb.com/title/tt1139797	A med student with a supernatural gift tries t...	205926.0	https://occ-0-4708-64.1.nflxso.net/dnm/api/v6/...	https://m.media-amazon.com/images/M/MV5BOWM4NT...	https://www.youtube.com/watch?v=LqB6XJix-dM
Caitlin Moran	https://www.netflix.com/watch/81041267	https://www.imdb.com/title/tt4193072	When nerdy Johanna moves to London, things get...	2838.0	https://occ-0-1081-999.1.nflxso.net/dnm/api/v6/...	https://m.media-amazon.com/images/M/MV5BZGUyN2...	https://www.youtube.com/watch?v=elboxPy4okQ
Pattaranad Bhiboonsawade, Mez Tharatorn, Thods...	https://www.netflix.com/watch/81306155	https://www.imdb.com/title/tt13393728	After her ex-boyfriend cons her out of a large...	131.0	https://occ-0-2188-64.1.nflxso.net/dnm/api/v6/...	https://m.media-amazon.com/images/M/MV5BODAzOG...	https://www.youtube.com/watch?v=md3CmFLGK8Y
NaN	https://www.netflix.com/watch/81307527	https://www.imdb.com/title/tt2300049	A group of social welfare workers led by their...	47.0	https://occ-0-2508-2706.1.nflxso.net/dnm/api/v6/...	https://m.media-amazon.com/images/M/MV5BMTc0Nz...	https://www.youtube.com/watch?v=5kyF2vy63r0
Ivar Lo-Johansson	https://www.netflix.com/watch/81382068	https://www.imdb.com/title/tt0041155	An unhappily married farm worker struggling to...	88.0	https://occ-0-2851-41.1.nflxso.net/dnm/api/v6/...	https://m.media-amazon.com/images/M/MV5BMjVzMz...	https://www.youtube.com/watch?v=H0itWKFWmPQ

Appendix D: Netflix Released dates vs. no. of content

